Sales and Inventory Analytics for Precast RCC Products

A Mid-Term report for the BDM capstone Project

Submitted by

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Executive Summary

Laxmi Concrete Products is a well-established trading firm specializing in Precast RCC products. Located in Village Sunped, Ballabgarh, the business serves both retail and wholesale customers with a range of RCC products such as benches, slabs, pipes etc. However, unpredictable demand and inconsistent cashflow due to credit sales affects profitability and growth.

This report aims to tackle these challenges, by analysing the past sales and inventory data, sourced directly through business records. Through effective descriptive and trend analysis, fluctuations in demand and inventory levels have been identified. A predictive forecasting of sales has been done to optimize the inventory management, ensuring stock levels are balanced with seasonal demands. Study will incorporate the credit sales analysis for reducing the risk of defaults and ensuring consistent cash flow.

The analysis helps to find the crucial relations between sales and inventory level, highlighting the need of better stock management techniques. All this help us to formulate the recommendations for the Laxmi Concrete Products to enhance the efficiency and have sustainable growth and finances.

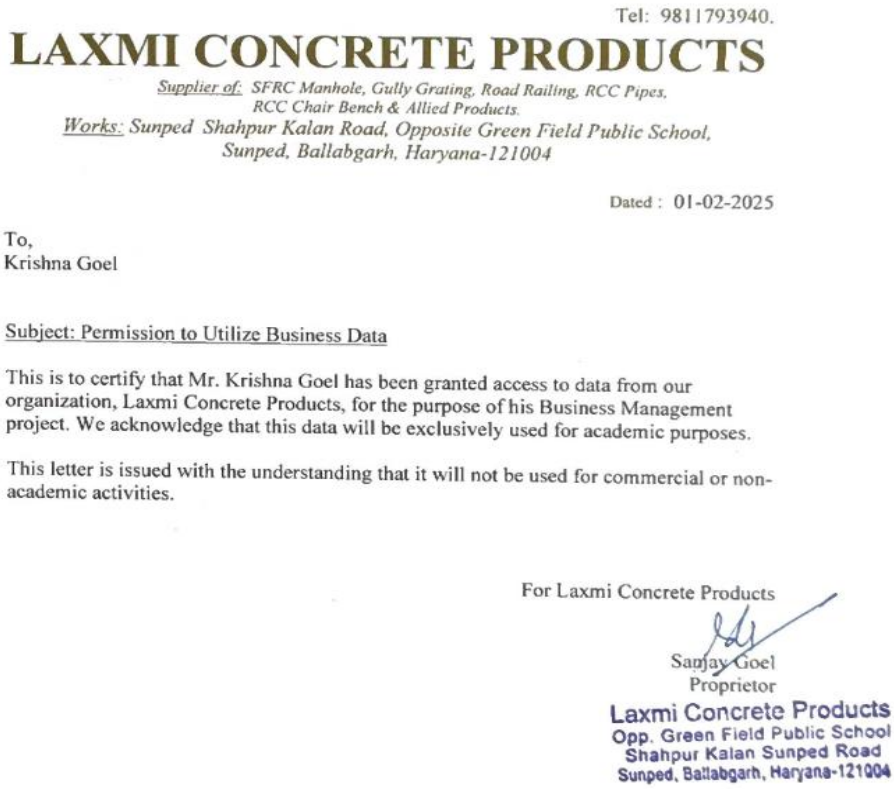
Proof of originality of the Data

Drive link for Proof of Originality of data:
https://drive.google.com/drive/folders/195NDNO_QyzxtzmwBjO9wsnmcRzw78lb2?usp=drive_link

Figure 1 Me at Laxmi Concrete Products Office

Figure 2: Interaction with owner

Interaction Video link:
https://drive.google.com/file/d/1_bTF2fFxCHjpi2mj8m5lBxNfGxigz_Xf/view?usp=drive_link



Tel: 9811793940.

LAXMI CONCRETE PRODUCTS

Supplier of: SFRC Manhole, Gully Grating, Road Railing, RCC Pipes, RCC Chair Bench & Allied Products.

Works: Sunped Shahpur Kalan Road, Opposite Green Field Public School, Sunped, Ballabgarh, Haryana-121004

Dated : 01-02-2025

To,
Krishna Goel

Subject: Permission to Utilize Business Data

This is to certify that Mr. Krishna Goel has been granted access to data from our organization, Laxmi Concrete Products, for the purpose of his Business Management project. We acknowledge that this data will be exclusively used for academic purposes.

This letter is issued with the understanding that it will not be used for commercial or non-academic activities.

For Laxmi Concrete Products

Sanjay Goel
Proprietor

Laxmi Concrete Products
Opp. Green Field Public School
Shahpur Kalan Sunped Road
Sunped, Ballabgarh, Haryana-121004

Figure 3 Letter from Organisation

Metadata & Descriptive Statistics

For this project the data was sourced from Laxmi Concrete Products. The data was collected in offline meetings through ledger entries and business records covering sales and inventory data from September 2024 to February 2025. The hand-noted data was collected and digitized by entering in excel sheet. To streamline analysis, each SKU was entered in a different sheet, making it more suitable to analyse sales and inventory of individual SKU.

The data collected was structured in following columns.

- **Date:** This represents specific day when any sales or purchase happened. It is very important for time series analysis of data for forecasting of sales.
- **SKU Name:** Used to identify each product uniquely.

- **Inwards:** This consist of opening stock of a SKU on a day and any purchase of SKU.
- **Outwards:** This consists of sales of a SKU on a day and value/Revenue for the SKU.
- **Closing:** This column represents the closing stock for a SKU on a particular day.

The excel tools like formulas, tables charts, descriptive statistics were employed to discover the sales and inventory pattern for SKUs. This helped to find out the demands and inventory level of SKUs.

DATASET LINK:
https://1drv.ms/x/c/7beb6c6285eab2e7/EVu6ALboyDlOvYceYt17CMcBh2lzOPKXnCExgLVwJFWlOg?e=CTgcJl

Descriptive Statistics

The descriptive statistics table provides key insights into sales distribution of different SKU.

SKUs	Mean	Median	Mode	Variance	Standard Deviation	Range	Count
HD-20	20	25	25	63	8	29	66
EHD-35	21	25	25	53	7	26	66
MD-10	17	10	10	137	12	32	7
Slab 1200*450*100mm	14	12	30	100	10	26	9
Slab 1000X500X70 mm	45	44	NA	225	15	30	3
Slab 1000X 500X100 mm	40	50	50	400	20	40	4
PIPE 300 NP2	15	15	NA	NA	NA	0	1
PIPE 250 NP2	12	12	NA	NA	NA	0	1
BENCH	8	10	10	14	4	9	9
BOLLARDS	216	200	350	23366	153	425	10

Each statistical measure plays important role in understanding the sales behaviour, demand variability and inventory management. Mean represents average unit sold for each SKU. Higher mean value indicates higher average sales like bollards have mean 216 and lower mean value indicate low average sales like benches. Median represents middle value of data, slab 1000x500x100mm has median of 50 units suggesting that half of observations lie below this. Mode helps to identify most frequent occurring sales figure, this is useful to detect common demand levels such as slab 1000x500x70mm has no mode, it indicates that the sales figures does not repeat. Variance quantifies spread of sales data; bollards have variance 23366 it shows extreme variability in demand. Range represents difference between highest and lowest sales, example bollards have range 425, it represents sharp changes in sales. Count represents number of observations recorded for reach SKU, SKUs like pipe have count 1 it represents that it was sold only one time over the six months.

Detailed Explanation of Analysis Process and Method

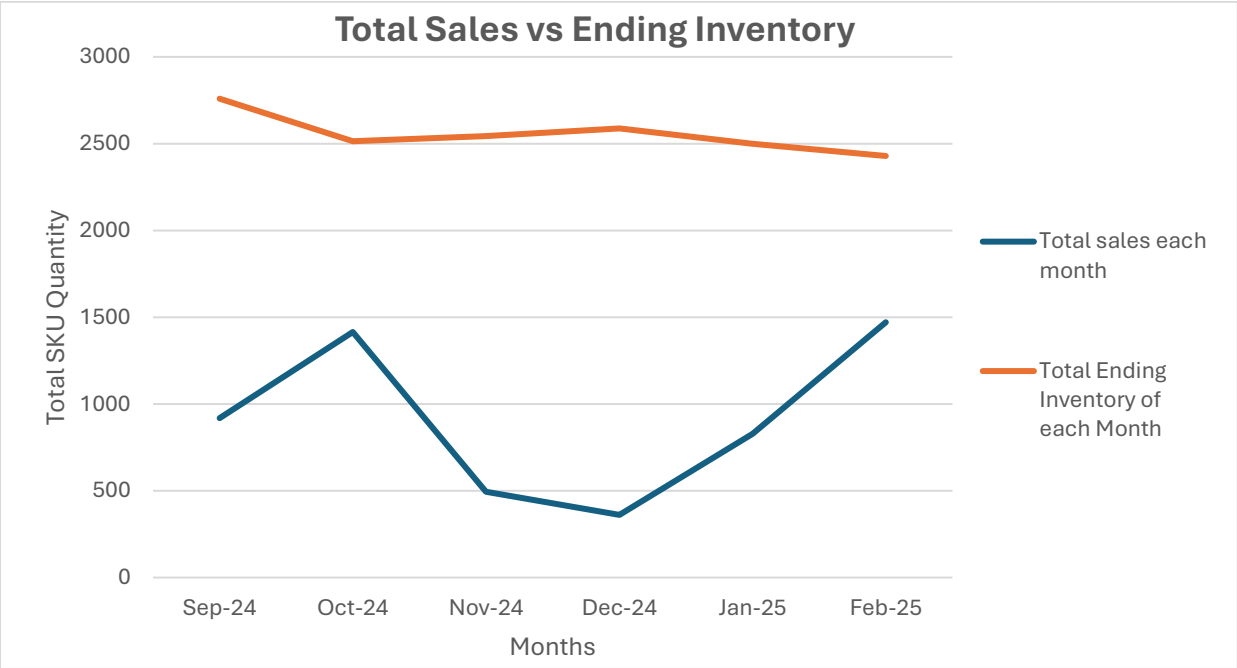
The analysis process of the project combines quantitative and qualitative methods. To handle the data Ms Excel is used. The data was entered in Excel and various functions were used to compute monthly sales and monthly ending inventory level of individual SKUs.

The functions and tools such as sum, lookup and data analysis tools were used simplify the calculations and perform descriptive statistics on the data. Charts are used to visualise pattern and trends, also it made the data easy to understand and summarise. This method is more efficient, practical and easy to perform and further reduces the need for complex tools.

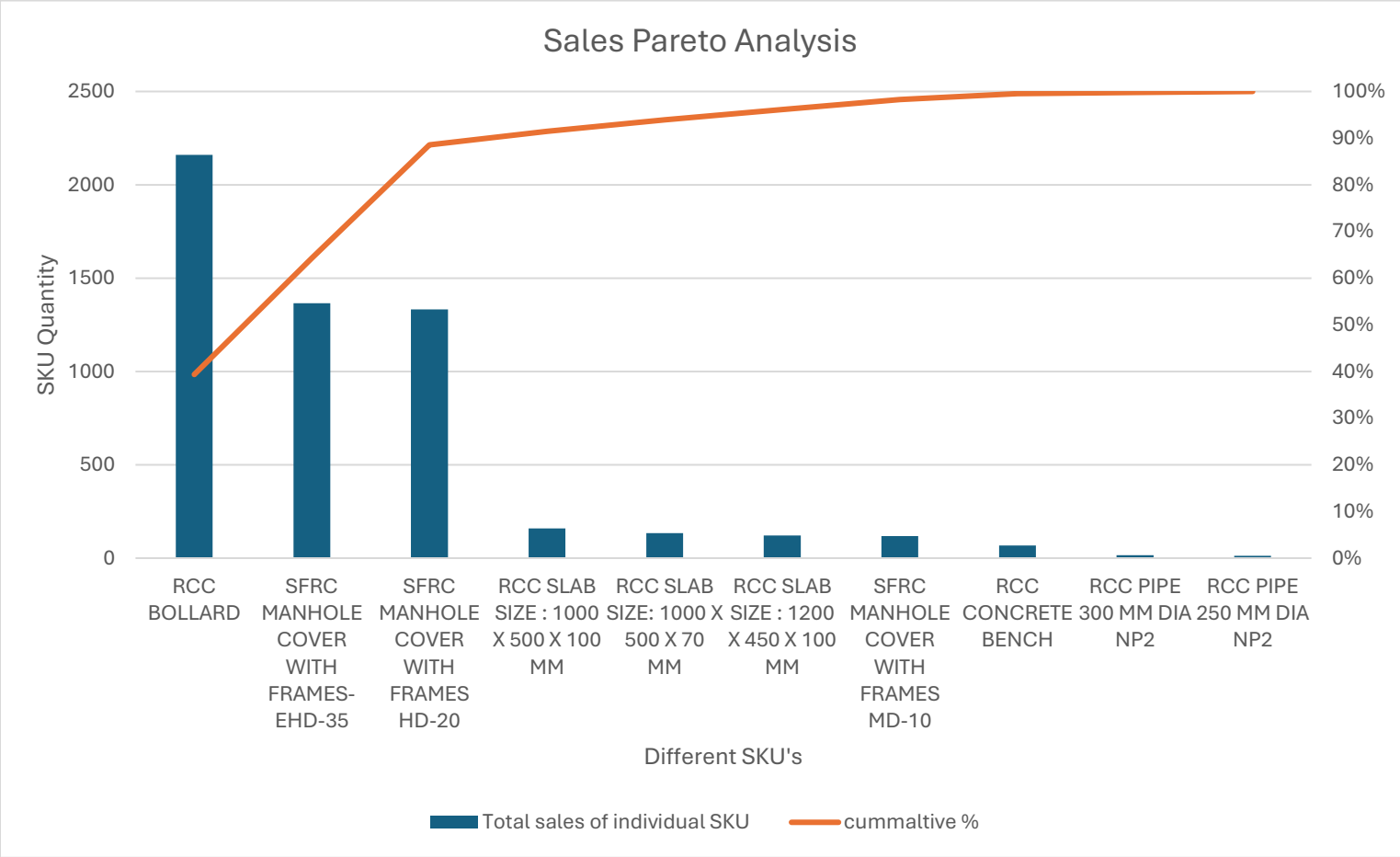
For Unpredictable Demand, Trend Analysis is used to analyse the last six-month sales data. This analysis helped to identify the past demand for different SKUs. Also, Trend analysis is performed on inventory data, which helped to correlate the monthly demand with monthly inventory level. This approach helped to further forecast the total demand using forecasting tools provided by Ms Excel and would also help to build an optimal balance between Demand and Inventory level. This method is more interpretable, accessible than other methods like machine learning as it is well suited for a small-scale business and easy to understand and perform.

By combining the different tools of Excel for calculations, visualisation and forecasting with analytical technique like Time Series Analysis of demand and inventory levels; valuable insights of demand and inventory pattern are obtained which will further help to make better decisions.

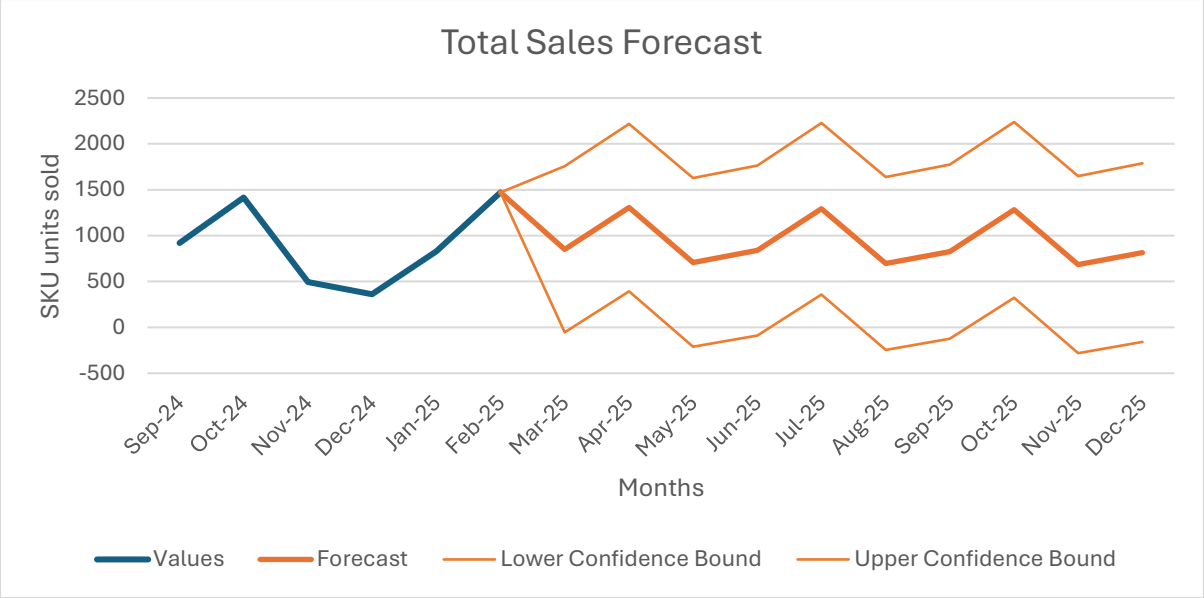
Results and Findings



The above line chart represents Total Sales vs Ending Inventory of SKUs over six months period. The sales data reveals fluctuations in demand throughout the months with peaks in October with total sales 1415 units and February with 1471 units. The lowest sales occur in December (361 units) highlighting seasonal variation in sales pattern. Despite of these sales variation the inventory levels remain stable ranging between 2759 units in October to 2429 units in February. This shows lack of alignment between inventory management and actual demand that led to inefficiency and increasing holding cost.



The analysis of SKU demand validates the Pareto Principle, showing that the top performing SKUs contribute to majority of sales. RCC Bollards, SFRC Manhole EHD-35 and HD-20 are top performing SKUs, these require priority inventory planning to prevent stockouts and ensure uninterrupted sales. The SKUs like RCC Pipe 300mm Dia and 250 mm Dia have very low sales, indicating minimal demand in market. Stocking excess quantity of these may lead to overstocking and increased holding costs.



The total sales forecast highlights significant fluctuations in demand throughout year, with peaks in April, July and October similar to past data. The months like May and August and November indicate the period where inventory need to be managed carefully. Confidence level indicates potential deviation in sales with possible drop in May and August or possible increase in April and October. If past trend continues then February and October will remain high sales months requiring proactive overstocking, while May and August may require promotions and more marketing to boost sales.

In final submission, I would find solutions to these problems by forecasting the demand of individual SKU and credit analysis.